MGMTMSA401: Machine Learning for Decision Making
Fall 2026

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Class Sessions:
Section 1: Wednesdays, 8:30am to 11:20am, Room G419
Section 2: Wednesdays, 1pm to 3:50pm, Room G419

1 Overview

Uncertainty permeates every domain. For example, in healthcare, for a given set of patients, we do not know which patient will experience an acute event; in finance, given a set of assets, we do not know which ones will experience a significant increase in their price; and in retail, we do not know how much demand there will be for a given product next week. Uncertainty is what makes decision making difficult. Without uncertainty, it would be trivial to make good decisions: we would know exactly which patient to prioritize for an intervention, which asset to invest our money into, and how much of a product to stock to meet demand.

To mitigate the impact of uncertainty, and to help us make good decisions, we can pursue the goal of making predictions about this uncertainty. In the healthcare setting, we could predict for each patient their probability of experiencing the acute event; in the finance example, we could predict how much each asset's price will change; and in the retail setting, we could predict how much demand there will be for a product. These predictions in general can be made with the help of data. For example, we might have data on patient characteristics in the healthcare setting, data on previous prices of the assets in the finance setting, and data on social media interest for the product we are stocking.

Machine learning is the science of using data to create models for making effective predictions. In this course, we will introduce the fundamental concepts relating to the construction of predictive models, and analysis of their quality and complexity. Our five lectures will develop these concepts and introduce the most commonly used classes of supervised learning models (linear regression, logistic regression, regularized regression, tree and tree ensemble models, and neural networks).

2 Course Materials, Assessment and Rules

2.1 Textbook

There is no required textbook for the course. Slides and any other relevant material will be made available through BruinLearn.

The following books may be helpful:

- Bertsimas, O'Hair and Pulleyblank, *The Analytics Edge*
  Amazon link: `https://tinyurl.com/yc6dbemw`
- James, Witten, Hastie, Tibshirani and Taylor, *An Introduction to Statistical Learning with Applications in R*
  Amazon link: `https://tinyurl.com/3yjxramb`
  Freely available at `https://tinyurl.com/43ukuf59`
- Géron, *Hands-On Machine Learning with Scikit-Learn and PyTorch*
  Amazon link: `https://tinyurl.com/2kvxv6e4`
- Chollet and Watson, *Deep Learning with Python*
  Amazon link: `https://tinyurl.com/5yvx4csx`

2.2 Grading

Your course grade will be determined as follows:

Class Participation	10%
Homework	30%
Final exam	60%

We will follow the standard curve in core classes:

Grade	**Frequency**
A	20%
A-	25%
B+	25%
B	25%
B- or lower	5%

A note on course grades: Canvas / BruinLearn may show an "overall grade" (e.g., 90.36% out of 100%) as well as a letter grade (e.g., A, B+, A-, etc.). Please disregard these, as these do not take into account all of the components of your grade and do not follow our overall curve.

2.3 Attendance, lateness and classroom etiquette

Attendance in class is required. I will take attendance at the start of each class. You are allowed to miss one class during the quarter without penalty. Missing more than one class will result in a reduction of your participation grade.

With regard to classroom etiquette, I expect you to do your best to follow along in class, and to be engaged in classroom discussions. If we are doing an exercise that involves coding, I expect your laptop to be open to the Jupyter Notebook we are working on (i.e., you should not use it as an opportunity to reply to emails, edit your resume, etc.). Do not go on your phone during class. Do not talk during class unless I have instructed you to discuss a problem with the person sitting next to you.

I welcome questions during class on the material, and I am happy to stop and explain something again or delve into further details. I do ask that you keep your questions focused on the material we are currently discussing. If you have questions on material we have covered in earlier lectures, please see me during the break or after class, or send me or the TA an email with your questions after class.

Lastly, being late to class is not acceptable. Students who enter the classroom while I am presenting are extremely disruptive both to myself and to the students. **Anyone who arrives late – while I am speaking or while another student is asking me a question – will receive a participation grade of zero out of ten.** If you find that you have arrived late, wait until the break to enter the classroom.

2.4 Office Hours

If you need help, I am happy to set up a meeting with you, subject to my availability. Aside from office hours, I am always happy to answer questions over email. Please always feel free to reach out to me, even if you think your question may be basic.

2.5 Homework

We will have two homework assignments. Each assignment will consist of several questions, each focusing on a data set, with multiple parts. You will be provided with the questions and the data sets.

Homework is due via BruinLearn by midnight of the day of class. (For example, Homework 1 will be due by midnight on November 18.) Your submission should include your answers to the problems as well as a (**short**) printout of the Python code you used to solve the problems. Each of the two homeworks is worth 15% of your overall grade.

Grades for late homework submissions are reduced by 25% for each day late, rounding up to the next day late. For example, an assignment that is received two hours late after class and receives a score of 80 out of 100 will receive an overall grade of $80/100 \times 0.75 = 60/100$. An assignment received four days or later than the deadline will receive an overall grade of 0/100.

You are allowed to discuss the problems with others, but **the assignment that you submit must be your own. Copying somebody else's Python code or their answers is a violation of academic integrity and will result in disciplinary action.**

2.6 Final exam

The final exam will be an in-person, closed book, closed notes exam. You are allowed a simple, non-graphing calculator. The exam time for both Sections 1 and 2 is **Wednesday, December 9**, **1pm - 4pm PST**. Additional details on the exam (room assignments, what material is covered, what types of problems to expect, etc.) will be shared closer to the date of the exam.

2.7 Laptop use, software and AI policy

We will be using Python in every class. Please bring your laptop to class and ensure that you are ready to use Python.

You are allowed to use artificial intelligence (AI) tools such as ChatGPT, Claud, etc. to help you solve problems on the homework. However, I would **strongly** discourage you from doing so, especially without struggling with the questions first; it is in that struggle that you truly learn what the concepts mean. Using AI obviates that struggle and inhibits learning and long-term retention of material. Note that the use of AI is *not* permitted on the final exam.

2.8 Regrading

If you believe that an assignment was not graded properly, please contact the TA and ask for a regrade. Clerical errors such as addition mistakes will be corrected without any risk. All other regrade requests will involve a complete re-grade of all questions. Students who request a regrade acknowledge that their grade may increase, stay the same or decrease. The result of the regrade is final. All regrades must be requested within one week of the distribution of grades for the given assignment/exam.

2.9 Extensions

In general, we do not grant extensions for homework assignments. Except in exceptional circumstances (e.g., imminent childbirth for self or spouse), you are expected to submit homework assignments by their designated deadlines.

In the extraordinary case that you are granted an extension for a homework assignment, we enforce a strict policy of a maximum of one extension per student: if you are granted an extension, you will not be granted any further extensions, either for the assignment in question, for any later assignments or for the final exam. If you do not submit an assignment by the extended deadline granted to you, the late penalty as described above under "Homework" will be applied relative to the extended deadline.

3 Schedule

(Note that the topic assigned to each week is approximate, and may change as we progress through the quarter.)

Class	**Date**	**Topic**	**Deliverables**
1	Nov 4	Linear Regression Logistic Regression	
2	Nov 13	Regularized Regression Intro to Tree Models	
3	Nov 18*	Tree Models	Homework 1 (due Nov 18 11:59pm PST for Sec 1 & 2)
4	Nov 25**	Neural Networks - I	
5	Dec 2	Neural Networks - II	Homework 2 (due Dec 2 11:59pm PST for Sec 1 & 2)
	Dec 9	**Final exam**	1pm - 4pm PST (Sec 1 & 2) Rooms: TBA

* Note that November 11 is Veteran's Day, and we will not meet on that day. Instead, both sections will meet on Friday November 13 at their usual times.

** Note that November 25 is the day immediately before Thanksgiving. For this day, the morning lecture from 8:30am to 11:20am will be held in person, while the afternoon lecture from 1pm to 3:50pm will be held on Zoom. A link for the Zoom session will be provided closer to the date of that class.